

Contents

- Boolean Algebra
 - Basic Definitions
 - Boolean Algebra: Postulates
 - Axiomatic Definition of Boolean Algebra
 - Binary Information Processing
 - Two-Valued Boolean Algebra
 - Differences between Boolean algebra and ordinary algebra
 - Duality Principle
 - Basic Theorems
 - Operator Precedence
- Binary Logic
 - AND Operation
 - OR Operation
 - NOT Operation
 - Logic gates

Boolean Algebra

❖ Basic Definitions

- Named after the nineteenth-century mathematician George Boole, Boolean logic is a form of algebra in which all values are reduced to either TRUE or FALSE.
- Boolean logic is especially important for computer science because it fits nicely with the binary numbering system, in which each bit has a value of either 1 or 0.
- Boolean Algebra is used to analyze and simplify the digital (logic) circuits.
- It is also called as Binary Algebra or logical Algebra.

Boolean Algebra

❖ Basic Definitions: Postulates

❖ The most common postulates used to formulate various algebraic structures are:

1. **Closure:** A set S is closed with respect to a binary operator if, for every pair of elements of S , the binary operator specifies a rule for obtaining a unique element of S .

For example, the set of natural numbers $N = \{1, 2, 3, 4, c\}$ is closed with respect to the binary operator $+$ by the rules of arithmetic addition, since, for any $a, b \in N$, there is a unique $c \in N$ such that $a + b = c$.

The set of natural numbers is not closed with respect to the binary operator $-$ by the rules of arithmetic subtraction,

Boolean Algebra

❖ Basic Definitions: Postulates

2. Associative Law: A binary operator $*$ on a set S is said to be associative whenever

$$(x*y)*z = x*(y*z) \quad \text{for all } x, y, z \in S$$

3. Commutative Law: A binary operator $*$ on a set S is said to be associative whenever;

$$x*y = y*x \quad \text{for all } x, y \in S$$

Boolean Algebra

❖ Basic Definitions: Postulates

4. Identity Element: A set S is said to have an identity element with respect to a binary operator $*$ on S if there exists an element $e \in S$ with property

$$x * e = e * x = x \quad \text{for every } x \in S$$

Example: The element 0 is an identity element with respect to the binary operator $+$ on the set of integers $I = \{ \dots, -3, -2, -1, 0, 1, 2, 3, \dots \}$, since $x + 0 = 0 + x = x$ for any $x \in I$

Boolean Algebra

❖ Basic Definitions: Postulates

5. Inverse: A set S having the identity element e with respect to a binary operator $*$ is said to have an inverse whenever, for every $x \in S$, there exists an element $y \in S$, such that

$$x * y = e$$

Example: In the set of integers, I , and the operator $+$, with $e = 0$, the inverse of an element a is $(-a)$, since $a + (-a) = 0$.

6. Distributive Law: If $+$ and $\cdot$ are two binary operator on a set S , $*$ is said to be distributive over whenever

$$x + (y \cdot z) = (x + y) \cdot (x + z)$$

&

$$x \cdot (y + z) = xy + xz$$

Boolean Algebra

❖ Axiomatic Definition of Boolean Algebra

1. The set B contains at least two elements $\{a, b\}$ with $a \neq b$

2. Closure: $a+b$ is in B $a \cdot b$ is in B

3. Commutative: $a+b = b+a$ $a \cdot b = b \cdot a$

4. Associative: $a+(b+c) = (a+b)+c$ $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

5. Identity: $a+0 = a$ $a \cdot 1 = a$

6. Distributive: $a+(b \cdot c) = (a+b) \cdot (a+c)$ $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$

7. Complementarity: $a+a' = 1$ $a \cdot a' = 0$

Boolean Algebra

❖ The differences between Boolean algebra and ordinary algebra:

- i. The distributive law of $+$ over $\cdot$, i.e., $x+(y \cdot z) = (x +y) \cdot (x +z)$, is valid for Boolean algebra, but not for ordinary algebra.
- ii. Boolean algebra does not have additive or multiplicative inverse; therefore, there are no subtraction or division operations.
- iii. Complement is not available in ordinary algebra.
- iv. Boolean algebra deals with the as yet undefined set of elements, B , but in the two-valued Boolean algebra, B is defined as a set with only two elements, 0 and 1.

Boolean Algebra

❖ Two-Valued Boolean Algebra

- The rule of a Two-Valued Boolean Algebra is shown in the following table:
- These rules are exactly the same as the AND, OR and NOT operations.
- A two-valued Boolean algebra is defined on a set of two elements, $B = \{0, 1\}$, with rules for the two binary operators $+$ and $\cdot$.

x	y	$x \cdot y$
0	0	0
0	1	0
1	0	0
1	1	1

x	y	$x + y$
0	0	0
0	1	1
1	0	1
1	1	1

x	x'
0	1
1	0

Boolean Algebra

❖ The distributive law for Two-valued Boolean

- According to the distributive law we can write;

$$x \cdot (y + z) = (x \cdot y) + (x \cdot z)$$

For each combination, we derive $x \cdot (y + z)$ and show that the value is the same as the value of $(x \cdot y) + (x \cdot z)$:

x	y	z
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

$y + z$	$x \cdot (y + z)$
0	0
1	0
1	0
1	0
0	0
1	1
1	1
1	1

$x \cdot y$	$x \cdot z$	$(x \cdot y) + (x \cdot z)$
0	0	0
0	0	0
0	0	0
0	0	0
0	0	0
0	1	1
1	0	1
1	1	1

Boolean Algebra

❖ Basic Theorems and Properties of Boolean Algebra

• Duality Principle

- Every algebraic expression deducible from the postulate of Boolean algebra remains valid if the operator and identity elements are interchanged.
 - replace “.” with “+”, and “+” with “.”;
 - replace “0” with “1”, and “1” with “0” ;
- If the dual of an algebraic expression is desired, we simply interchange **OR** and **AND** operators and replace 1's by 0's and 0's by 1's.

Boolean Algebra

❖ Basic Theorems

- Theorem 1(a): $x + x = x$
- Theorem 1(b): $x \cdot x = x$
- Theorem 2(a): $x + 1 = 1$
- Theorem 2(b): $x \cdot 0 = 0$
- Theorem 3: $(x')' = x$
- Theorem 4 (Associative): $(X+Y)+Z=X+(Y+Z)$
Dual: $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$
- Theorem 5 (De Morgan's): $(X+Y+\dots)' = X' \cdot Y' \cdot \dots$
Dual: $(X \cdot Y \cdot \dots)' = X' + Y' + \dots$
- Theorem 6(a): $x + xy = x$
- Theorem 6(b): $x(x + y) = x$
- Postulate 2 (Identity): $X + 0 = X$
- Dual: $X \cdot 1 = X$
- Postulate 4 (Distributive): $X \cdot (Y+Z) = (X \cdot Y) + (X \cdot Z)$
Dual: $X + (Y \cdot Z) = (X + Y) \cdot (X + Z)$
- Postulate 5 (Complementarity): $X + X' = 1$
- Dual: $X \cdot X' = 0$
- Postulate 3 (Commutative): $X + Y = Y + X$
- Dual: $X \cdot Y = Y \cdot X$

Boolean Algebra

❖ Proved of Theorem

- 01(a). **THEOREM 1(a):** $x + x = x$.

Statement	Justification
$x + x = (x + x) \cdot 1$	postulate 2(b)
$= (x + x)(x + x')$	5(a)
$= x + xx'$	4(b)
$= x + 0$	5(b)
$= x$	2(a)

- 01(b). **THEOREM 1(b):** $x \cdot x = x$.

Statement	Justification
$x \cdot x = xx + 0$	postulate 2(a)
$= xx + xx'$	5(b)
$= x(x + x')$	4(a)
$= x \cdot 1$	5(a)
$= x$	2(b)

Boolean Algebra

❖ Proved of Theorem

- 02(a). **THEOREM 2(a):** $x + 1 = 1$.

Statement	Justification
$x + 1 = 1 \cdot (x + 1)$	postulate 2(b)
$= (x + x')(x + 1)$	5(a)
$= x + x' \cdot 1$	4(b)
$= x + x'$	2(b)
$= 1$	5(a)

- 06(b). **THEOREM 6(a):** $x + xy = x$.

Statement	Justification
$x + xy = x \cdot 1 + xy$	postulate 2(b)
$= x(1 + y)$	4(a)
$= x(y + 1)$	3(a)
$= x \cdot 1$	2(a)
$= x$	2(b)

Boolean Algebra

❖ Operator Precedence

1. Parentheses

$$(\dots) \bullet (\dots)$$

2. NOT

$$x' + y$$

3. AND

$$x + x \bullet y$$

4. OR

Boolean Algebra

❖ Operator Precedence: DeMorgan's Theorem

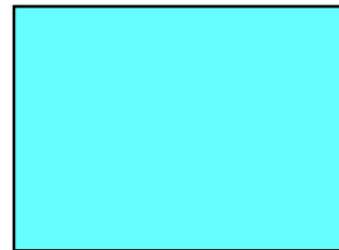
$$\begin{aligned}\overline{a [b + c (d + \bar{e})]} &= \bar{a} + \overline{[b + c (d + \bar{e})]} \\ &= \bar{a} + \bar{b} \overline{[c (d + \bar{e})]} \\ &= \bar{a} + \bar{b} (\bar{c} + \overline{[d + \bar{e}]}) \\ &= \bar{a} + \bar{b} (\bar{c} + (\bar{d} \bar{e})) \\ &= \bar{a} + \bar{b} (\bar{c} + \bar{d} e)\end{aligned}$$

Boolean Algebra

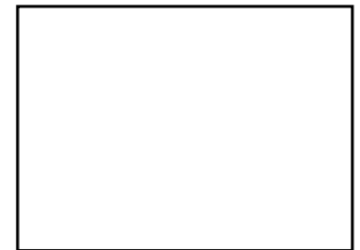
❖ Boolean Relationships on Venn Diagrams

- **Venn diagram:** graphical illustration of various operations and relations in an algebra of sets
- Elements of the set are represented by the area enclosed by a contour (usually a circle)

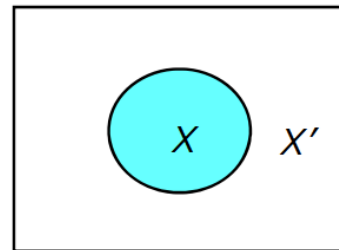
Venn diagrams for Boolean Algebra



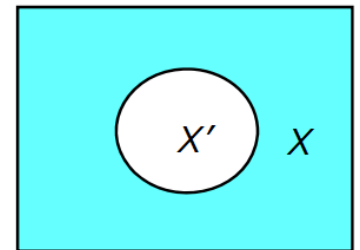
(a) Constant 1



(b) Constant 0



(c) Variable X



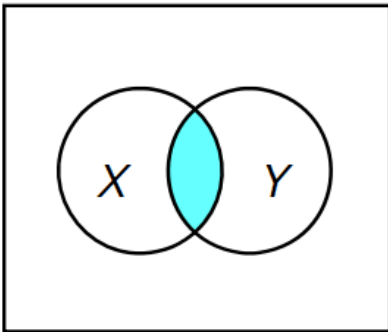
(d) X'

Boolean Algebra

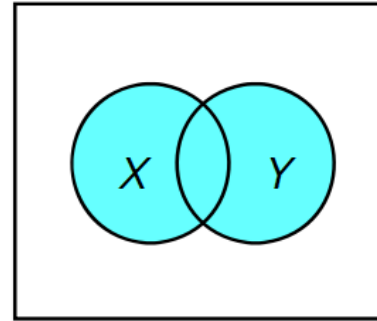
❖ Boolean Relationships on Venn Diagrams

- Venn Diagram illustration

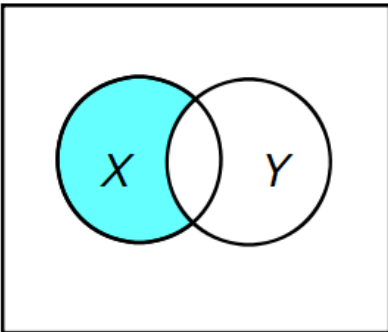
a) $F = xy$



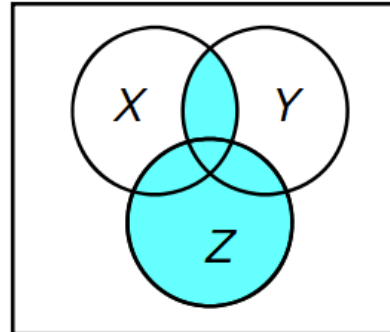
b) $F = x + y$



c) $F = xy'$



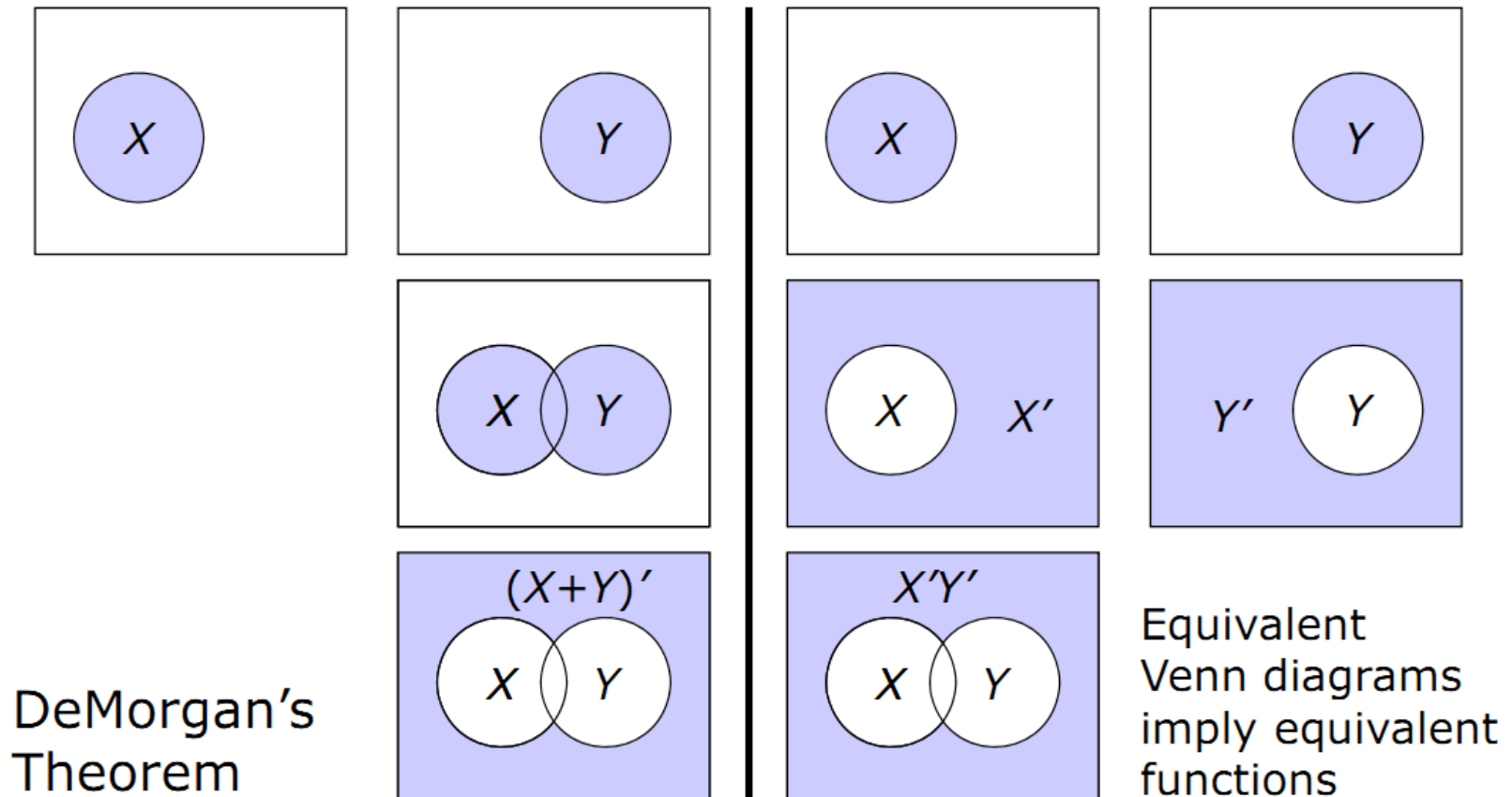
d) $F = xy + z$



Boolean Algebra

❖ Boolean Relationships on Venn Diagrams

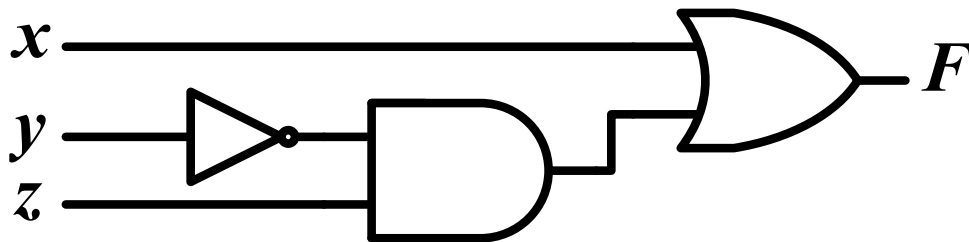
- Venn Diagram illustration: DeMorgan's Theorem: $F = (x+y)' = x'y'$



Boolean Algebra

❖ Boolean Functions

- Boolean Expression
 - Example: $F = x + y' z$
- Truth Table: All possible combinations of input variables
- Logic Circuit



Boolean Algebra

❖ Boolean Functions

- Example:

$$Z = A'BC + AB'C' + AB'C + ABC' + ABC$$

$$= A'BC + AB'(C' + C) + AB(C' + C) \quad \text{distributive}$$

$$= A'BC + AB' + AB \quad \text{complementary}$$

$$= A'BC + A(B' + B) \quad \text{distributive}$$

$$= A'BC + A \quad \text{complementary}$$

$$= BC + A \quad \text{absorption \#2 Duality}$$

$$(X \cdot Y') + Y = X + Y \text{ with } X=BC \text{ and } Y=A$$

Boolean Algebra

❖ Algebraic Manipulation

- ❖ **Literal:** A single variable within a term that may be complemented or not.
- **Use Boolean Algebra to simplify Boolean functions to produce simpler circuits**

Example: Simplify to a minimum number of literals

$$F = x + x' y \quad (\textcolor{blue}{3} \text{ Literals})$$

$$= x + (x' y)$$

Distributive law (+ over •)

$$= (x + x') (x + y)$$

$$= (1) (x + y) = \textcolor{red}{x} + \textcolor{red}{y} \quad (\textcolor{red}{2} \text{ Literals})$$

Tutorial Solve

Problem 01: Demonstrate by means of truth tables the validity of the following identities:

(a) DeMorgan's theorem for three variables: $(x+y+z)' = x'y'z'$ and $(xyz)' = x' + y' + z'$

(b) The distributive law: $x + yz = (x+y)(x+z)$

Solⁿ: (a)

x	y	z	$x+y+z$	$(x+y+z)'$
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	1	0
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0

x'	y'	z'	$x'y'z'$
1	1	1	1
1	1	0	0
1	0	1	0
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	0

x	y	z	xyz	$(xyz)'$
0	0	0	0	1
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

x'	y'	z'	$x' + y' + z'$
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	1
0	1	0	1
0	0	1	1
0	0	0	0

Tutorial Solve

Problem 01: Demonstrate by means of truth tables the validity of the following identities:

(a) DeMorgan's theorem for three variables: $(x+y+z)' = x'y'z'$ and $(xyz)' = x' + y' + z'$

(b) The distributive law: $x+yz = (x+y)(x+z)$

Solⁿ: (b)

			yz	x+yz
x	y	z		
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

x+y	x+z	(x+y)(x+z)
0	0	0
0	1	0
1	0	0
1	1	1
1	1	1
1	1	1
1	1	1
1	1	1

Tutorial Solve

Problem 02: Find the complement of $F = wx + yz$; then show that $FF' = 0$ and $F + F' = 1$.

Solⁿ:

Given, $F = (WX + YZ)$

Complement of F ;

$$\begin{aligned} F' &= (WX + YZ)' \\ &= (WX)'(YZ)' * (W' + X') \cdot (Y' + Z') \end{aligned}$$

$$\begin{aligned} F \cdot F' &= (WX + YZ) * (W' + X') * (Y' + Z') \\ &= (WX + YZ) * (W'Y' + W'Z' + X'Y' + X'Z') \\ &= (\cancel{W}\cancel{W}'X.Y' + \cancel{W}\cancel{W}'X.Z' + W\cancel{X}.X'Y' + W\cancel{X}.X'Z' + Z.W'\cancel{Y}\cancel{Y}' + YW'\cancel{Z}\cancel{Z}' + Z.X'\cancel{Y}\cancel{Y}' + Y.X'\cancel{Z}\cancel{Z}') \\ &= 0 \end{aligned}$$

Tutorial Solve

Problem 02: Find the complement of $F = wx + yz$; then show that $FF' = 0$ and $F + F' = 1$.

Solⁿ:

Given, $F = (WX + YZ)$

Now; Let, $(WX = A)$ and $(YZ = B)$

Then, $F = A + B$, $F' = (A + B)' = A' \cdot B'$,

$$F + F' = (A + B) + (A' \cdot B') = (A + A' \cdot B') + B$$

$$= (A + A')(A + B') + B$$

$$= (1)(A + B') + B$$

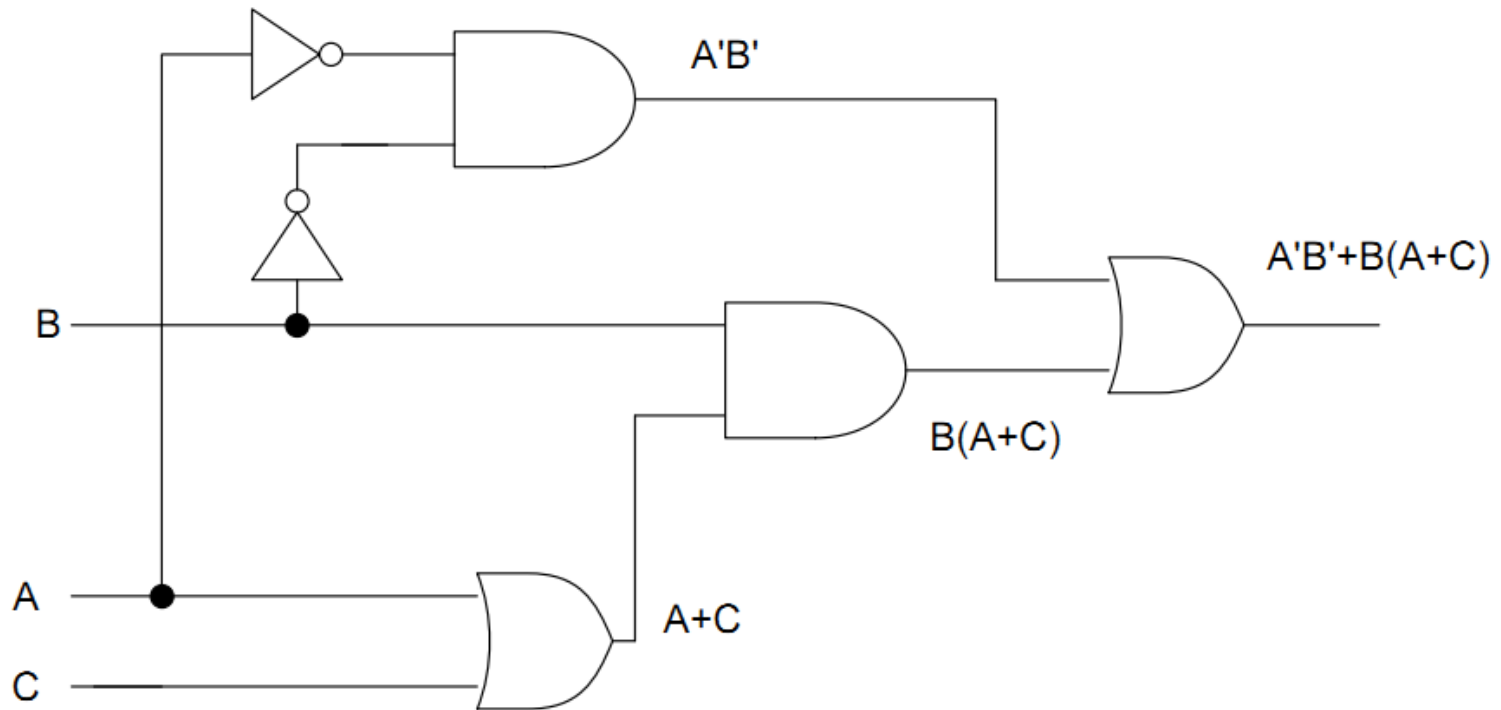
$$= A + B' + B = A + 1 = 1$$

Tutorial Solve

Problem 03: Draw the logical diagrams for the following Boolean expressions:

(a) $Y = A'B' + B(A+C)$.

Solⁿ:



Tutorial Exercise

Problem 01: Demonstrate the validity of the following identities by means of truth tables:

a) *The associative law:* $x + (y + z) = (x + y) + z$

b) *The distributive law:* $x(y + z) = xy + xz$

Problem 02: Find the complement of the following expressions:

a) $xy' + x'y$

b) $(a + c)(a + b')(a' + b + c')$

c) $z + z'(v'w + xy)$

Problem 03: Draw the logical diagrams for the following Boolean expressions:

a) $Y = (A + B)(C' + D)$

b) $Y = A'B(D' + C'D) + B(A + A'CD)$

Problem 02: Simplify the following Boolean expressions to a minimum number of literals:

$$F = (a + b + c')(a'b' + c)$$

Binary Logic

❖ Binary Logic

- Binary logic deals with variables that take on two discrete values and with operations that assume logical meaning.
- Binary logic consists of binary variables and a set of logical operations.
- The variables are designated by letters of the alphabet, such as A, B, C, x, y, z, etc, with each variable having two and only two distinct possible values: 1 and 0,
- Three basic logical operations: **AND, OR, and NOT**

Binary Logic

❖ AND Operation:

- This operation is represented by a dot or by the absence of an operator.

$$x \bullet y = z \quad \text{or, } xy = z \quad \text{or, } x \text{ AND } y = z$$

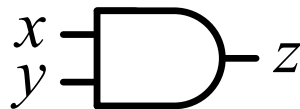
- The logical operation “**AND**” is interpreted to mean that,

$$z = \begin{cases} 1 & \text{if } x = 1 \text{ and } y = 1 \\ 0 & \text{otherwise} \end{cases}$$

- x , y , and z are binary variables and can be equal either to 1 or 0, and nothing else.

Boolean Expressions

$$z = x \bullet y = x y$$



Logic Gates

X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

Truth Tables

Binary Logic

❖ OR Operation:

- This operation is represented by a plus sign.

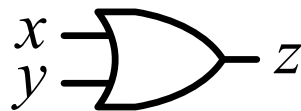
$$x + y = z \quad \text{or,} \quad x \text{ OR } y = z$$

- The logical operation “**OR**” is interpreted to mean that,

$$z = \begin{cases} 0 & \text{if } x = 0 \text{ and } y = 0 \\ 1 & \text{otherwise} \end{cases}$$

Boolean Expressions

$$z = x + y$$



Logic Gates

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Truth Tables

Binary Logic

❖ NOT Operation:

- This operation is represented by a prime (sometimes by an over-bar)

$$x' = z \quad \text{or,} \quad \bar{x} = z$$

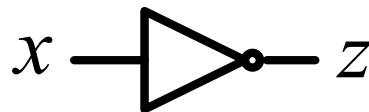
- The logical operation “**OR**” is interpreted to mean that,

$$z = \begin{cases} 0 & \text{if } x = 1 \\ 1 & \text{if } x = 0 \end{cases}$$

- The **NOT** operation is also referred to as the complement operation, since it changes a 1 to 0 and a 0 to 1

Boolean Expressions

$$z = x'$$



Logic Gates

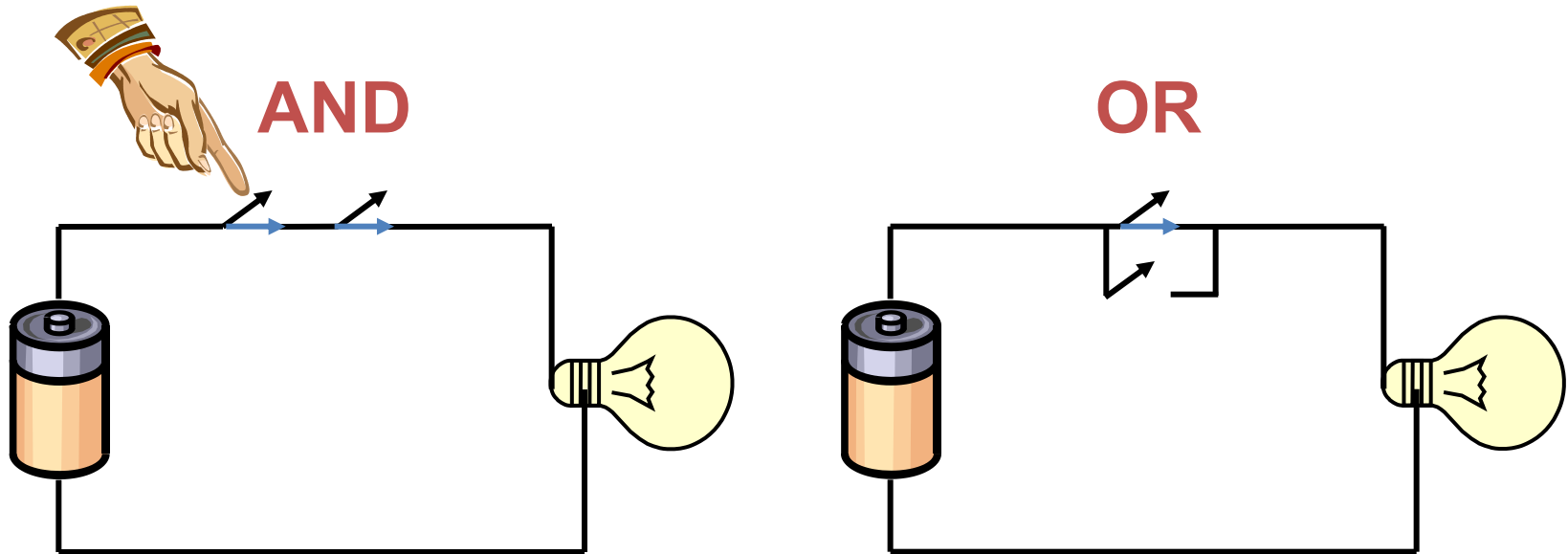
X	Z
0	1
1	0

Truth Tables

Binary Logic

Switching Circuits

- Binary logic should not be confused with binary arithmetic



Logic gates are electronic circuits that operate on one or more input signals to produce an output signal.

Binary Logic

❖ Binary Logic

- Logic gates
 - Example of binary signals

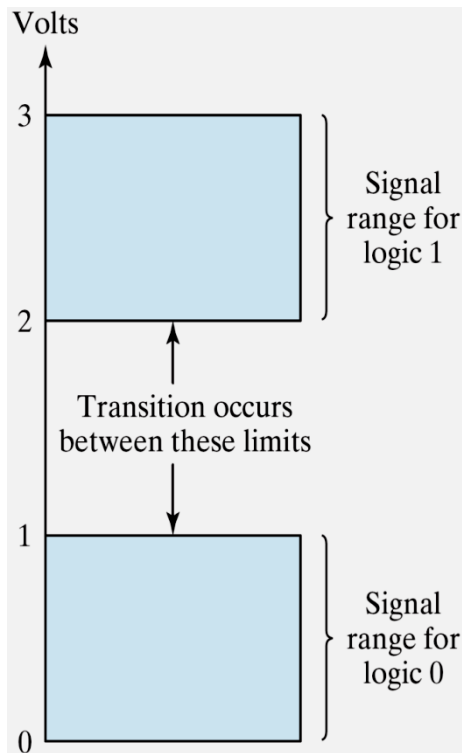
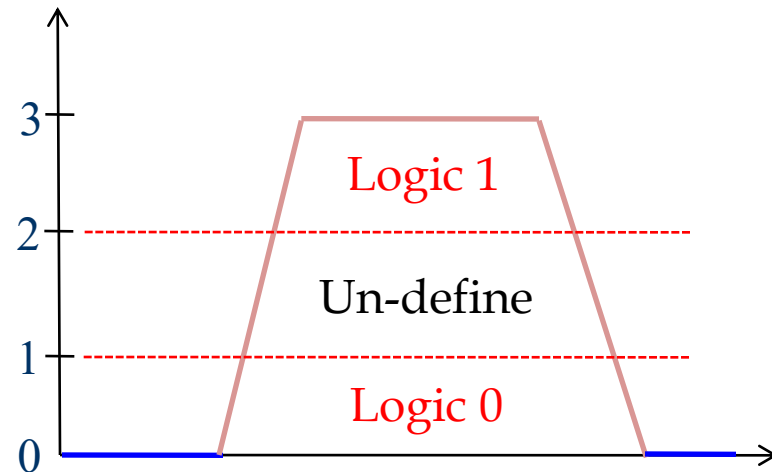


Figure: Example of binary signals

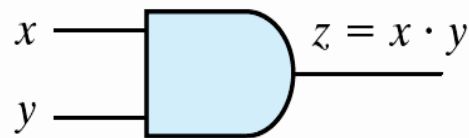


- The input terminals of digital circuits accept binary signals within the **allowable range** and respond at the output terminals with binary signals that **fall within the specified range**.
- The **intermediate region** between the allowed regions is crossed only during a state transition.
- When the physical signal is in a **particular range** it is interpreted to be either a 0 or a 1.

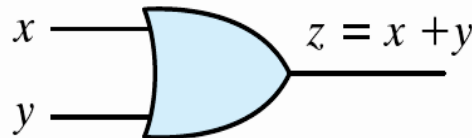
Binary Logic

❖ Logic gates

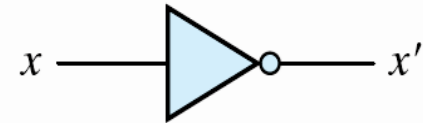
- Graphic Symbols and Input-Output Signals for Logic gates:



(a) Two-input AND gate



(b) Two-input OR gate



(c) NOT gate or inverter

Fig: Symbols for digital logic circuits

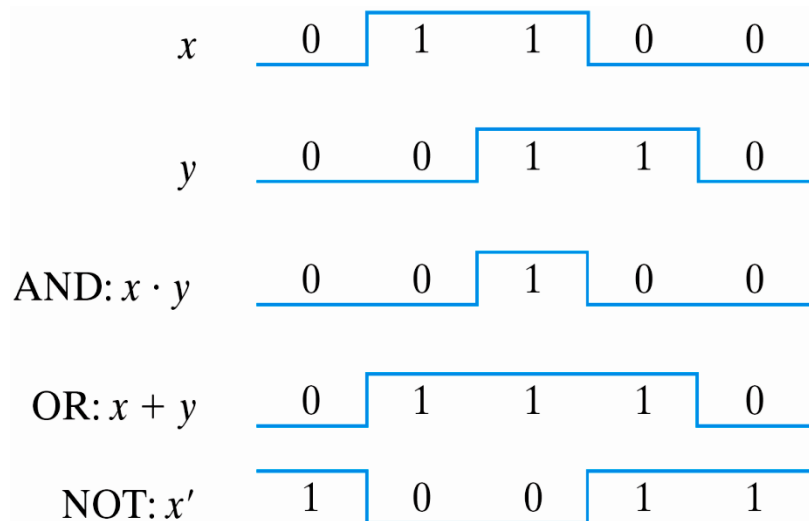


Fig: Input-Output signals for gates

- The gates are blocks of hardware that produce the equivalent of logic-1 or logic-0.
- The input signals x and y in the AND and OR gates may exist in one of four possible states: 00, 10, 11, or 01.
- The timing diagrams illustrate the idealized response of each gate to the four input signal combinations.

Binary Logic

❖ Logic gates

- Graphic Symbols and Input-Output Signals for Logic gates:

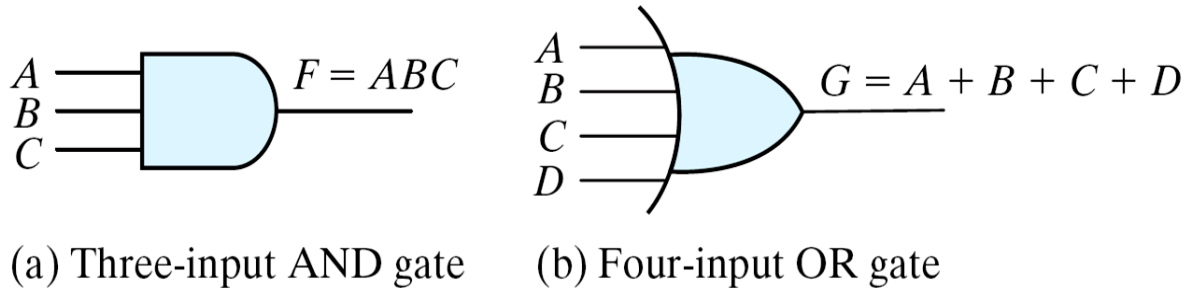


Fig: Gates with multiple inputs

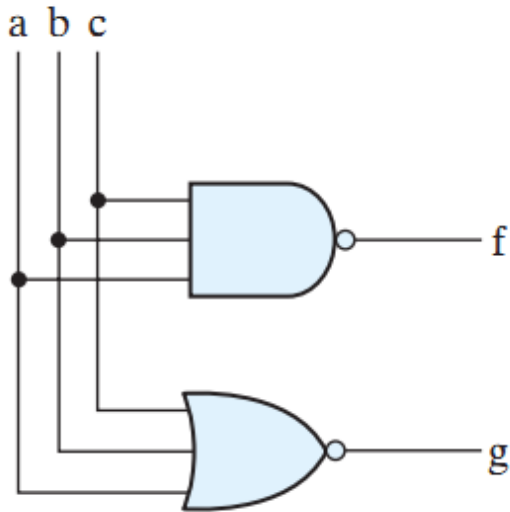
- AND and OR gates may have more than two inputs.
- The three-input **AND** gate responds with logic 1 output if all three inputs are logic 1. The output produces logic 0 if any input is logic 0.
- The four-input **OR** gate responds with logic 1 if any input is logic 1.
- In **OR** gate, output becomes logic 0 only when all inputs are logic 0.

Tutorial Solve

Problem 01: Show the signals of the outputs f and g in Fig. P1 as functions of the three inputs a , b , and c . Use all eight possible combinations of a , b , and c .

Use all eight possible combinations of a , b , and c .

000; 001; 010; 011; 100; 101; 110; 111



a	0	1	0	1	0	1	0	1	0	1	0	1	
b	0	0	1	1	0	0	1	1	0	0	1	1	
c	0	0	0	0	1	1	1	1	0	0	0	0	
f	1	1	1	1	1	1	1	0	1	1	1	1	
g	1	0	0	0	0	0	0	0	1	0	0	0	

Tutorial Solve

Problem 02: List the truth table of the functions:

a) $F = xy + xy' + y'z$

b) $F = bc + a'c'$

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

a	b	c	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

Tutorial Exercise

Problem 01: Show the signals of the outputs f and g in Fig. P1 as functions of the two inputs a and b . Use all four possible combinations of a and b .

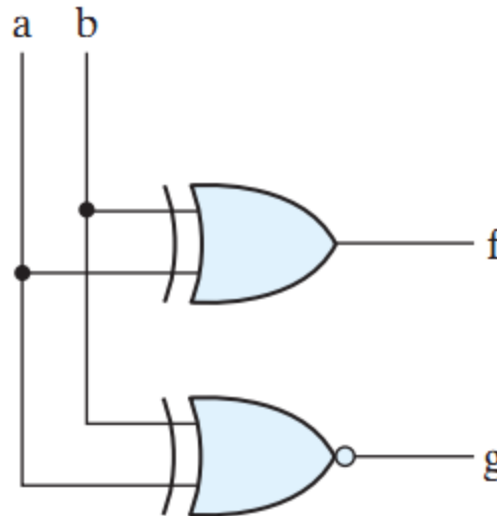


Fig. P1

Tutorial Exercise

Problem 02: Show the signals of the outputs f and g in the figure below as functions of the two inputs a and b . Use all four possible combinations of a and b

